

Overview

- ▶ **Goal**
 - ▶ Make use of Semantic Web technologies for a challenging problem
 - ▶ Apply approach to real data
- ▶ **Pass for mini-project** iff ...
 1. ROC AUC above 60% in GERBIL
 2. Documented code on GitHub/GitLab
 3. README with clear instructions for execution
- ▶ **Note:** Project not executable = **FAIL**



Organization

Mini-Project

Goal

Build a fact-checking engine, which returns a **veracity value** between 0 (fact is false) and +1 (fact is true) for a given **fact** with respect to a **knowledge graph**.

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- ▶ **Group size:** max. 3 persons
- ▶ **Code & documentation:** GitHub/GitLab
- ▶ **Suggested steps**
 1. Analyse the given training data
 2. Check how you can make use of the given knowledge base
 3. Fact Checking and Benchmarking
 4. Final submission: **July 15th, 2026**
- ▶ Do **not** wait for the lecture to cover this topic!

Mini-Project

Evaluation overview

Simple three steps to get an evaluation result.

1. TTL file in PANDA containing training data
(Will be uploaded during the next weeks)
2. Your approach should generate a result file
3. Result files can be uploaded in GERBIL for evaluation

Mini-Project

Training data

The input is represented as RDF statement

```
<http://swc2017.aksw.org/task2/dataset/3892429>  
  rdf:type rdf:Statement ;  
  rdf:subject <http://dbpedia.org/resource/Venus_Williams> ;  
  rdf:predicate <http://dbpedia.org/ontology/birthPlace> ;  
  rdf:object  
<http://dbpedia.org/resource/Lynwood,_California> .
```

Result format

Result file format (result.ttl)

- ▶ One line per fact
`<Fact-URI> <prop-URI> "value"^^type .`
- ▶ Fact-URI is the URI of the `rdf:Statement`
`http://swc2017.aksw.org/task2/dataset/3892429`
- ▶ the prop-URI is always
`http://swc2017.aksw.org/hasTruthValue`
- ▶ the value is the result of your fact checking algorithm (double)
- ▶ type of your value should be always
`<http://www.w3.org/2001/XMLSchema#double>`

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Possible result for the example:

```
<http://swc2017.aksw.org/task2/dataset/3892429>
  <http://swc2017.aksw.org/hasTruthValue>
    "0.8901"^^<http://www.w3.org/2001/XMLSchema#double> .
```

Mini-Project

Evaluate result

1. Go to <http://gerbil-kbc.aksw.org/gerbil/config>.
2. Choose "Fact Checking" as experiment type.
3. Enter your team name and a mail address.
The address won't be visible. It is used to make sure that you are part of this team.
4. Upload your result file.
5. Choose "FOKG 2024 Train" as reference dataset.
(Later, "FOKG 2024 Test" for the final evaluation.)
6. Agree to publish your result.
7. Agree to the disclaimer.
8. Submit your result.

Mini-Project

Evaluate result

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SWC 2019 Submission

Task ⓘ

Fact Checking ▾

Submission ⓘ

Upload a file with answers:

Name of participating system:E-mail address:

✖ My Team(My Email)(d346ceae-8241-4d5b-9c6f-ec4b28dc9e79_ result-test.ttl)

Or add a webservice via URI:

Name:

URI:

Reference dataset ⓘ

SW 2022 Test ▾

Or upload a dataset:

Name:

Publish ⓘ ☒

Disclaimer ☒ I agree to all data provided being stored and processed.
Any recourse to courts of law is excluded.

Mini-Project

Evaluate result

- ▶ GERBIL will determine the Area Under Curve (AUC) of your systems ROC curve. This value has the range $[0, 1]$.
- ▶ All results will be visible in a leaderboard which can be accessed at <http://gerbil-kbc.aksw.org/gerbil/overview>.
- ▶ If you see the message "*The annotator couldn't be loaded*" instead of a result your file has a wrong format (Checking it with an RDF validator might be a good way to go).
- ▶ Questions regarding the Mini-Project:
Umair Qudus: uqudus@mail.uni-paderborn.de and
Yasir Mahmood: ymahmood@mail.uni-paderborn.de

Mini-Project

Evaluate result

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About

GERBIL Experiment

Experiment URI: <http://gerbil-kbc.aks.w.org/gerbil/experiment?id=...> and <http://w3id.org/gerbil/kbc/experiment?id=...>

Type: Fact Checking

